

MINI PROJECT 01

DESIGNING A MODERN DATA PLATFORM

FOR ABC HUB

Data Warehouse Design Fundamentals

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1 Operational Systems Overview

Operational System	Description
Customer Management	Register and manage their accounts.
Content Management	Stores the catalogue of digital and physical content, metadata, availability information, ratings, reviews, and wish lists where applicable.
Rental Management	Records physical rental requests, rental dates, due dates, and returns.
Subscription Management	Records plans, enrolments, renewals, cancellations, and subscription status.
Payment Processing	Payment Processing
Inventory Management	Tracks physical copies, their availability, and stock status as items are rented and returned.
Customer Support	Records customer tickets, issues, and resolutions.
Delivery Management	Records dispatches, deliveries, returns in transit, and delivery status.

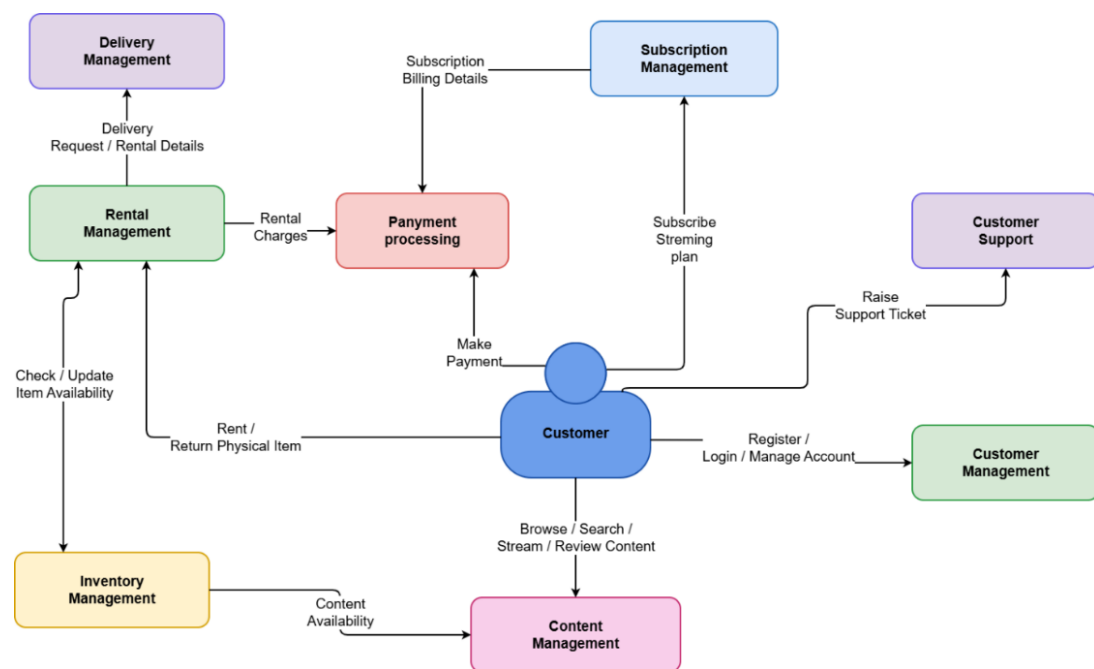


Figure 1 High level diagram of Operational systems interaction

2 Business Challenges and Proposed Solution

Challenges Discovered

- Business reports take a long time to prepare.
- Different departments often report different business figures.
- Historical business trends are difficult to analyse.
- Customer behavior across streaming and physical rentals cannot be analyzed together.
- Business users rely heavily on IT teams to prepare reports.
- International expansion is expected to increase both customer numbers and data volume significantly.

These challenges may arise partly because reporting data is spread across operational systems designed for day-to-day transactions. Preparing reports directly from those systems can require complex joins and aggregations. Such queries may be slow and compete with normal transactional workloads.

Different departments may also use different definitions for the same business measure. For example, Customer Management may count all registered customers, while Subscription Management may count only customers with active plans. Without an agreed definition of an “active customer,” their figures will differ even if both reports are calculated correctly.

Customer information and activity are also stored across several systems, including Customer Management, Subscription Management, Rental Management, and Payment Processing. Streaming activity and physical rentals must be linked using consistent customer identifiers and agreed business rules before they can be analyzed together. Without this matching and standardization, cross-system reports may be incomplete or inaccurate.

Operational systems may overwrite values or retain only limited history, depending on how they are designed. This makes long-term trend analysis difficult unless the required historical records are captured and retained in an analytical platform.

Operational records are organized to support individual business transactions. For company-wide reporting, the data needs to be integrated, cleaned, and presented through clearly defined business measures. Providing prepared analytical datasets will reduce business users’ dependence on IT for routine reports.

Therefore, the solution should provide:

- A reporting environment separate from operational workloads.

- Historical data for trend analysis.
- Consistent customer and content identifiers across sources.
- Validated data and agreed definitions for business measures.
- Business-friendly datasets that support self-service reporting.
- Capacity to grow as the company expands internationally.

Recommended Solution

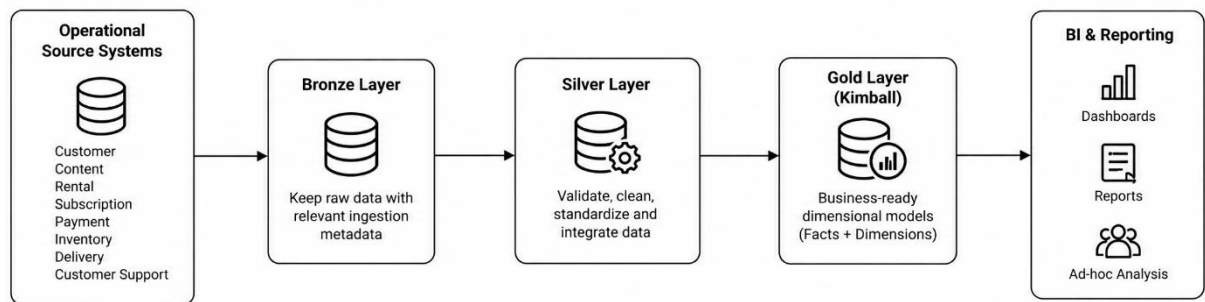


Figure 2 Proposed high-level architecture

I recommend a **data warehouse** for structured analytical data and reporting. A **Medallion-style pipeline** can organize the data into bronze, silver, and gold layers. Kimball-style dimensional modelling can be used in the gold layer to make reports easier to build and understand.

- **Bronze:** Ingest source data from operational systems and retain the source records and relevant ingestion metadata.
- **Silver:** Validate and clean records, reconcile customer and content identifiers, and standardize fields such as country codes, currencies, dates, and time zones.
- **Gold:** Publish dimensional data marts, agreed KPI definitions, and useful aggregates for business intelligence and self-service reporting.

For example, a shared definition of an active customer can make departmental figures consistent. Matching customer identifiers lets analysts examine streaming and rental behavior together. Retaining the required history enables trend analysis, while prepared Gold-layer datasets reduce the need for business users to request custom reports from IT.

The layers separate ingestion, integration, and reporting responsibilities. As data volume grows, appropriate partitioning, incremental processing, and aggregation can help maintain performance. For international revenue comparisons, the company must also agree on a reporting currency, an exchange-rate source, and a conversion rule; standardizing currency codes alone is insufficient.

3 Gathering Analytical Requirements

3.1 Customer Activity

Initial Requirement: - Management wants to analyses customer activity.

Requirement Area	Stakeholder Questions
Business Objective	• What business decisions should customer activity analysis support? • Do you want to measure customer engagement, customer retention, or both • Should streaming and physical rental activities be analyzed together?
KPIs and Measures	• Which activities should be considered customer activity? • Should we measure the number of streaming sessions? • Should we measure total streaming duration? • Should we measure the number of physical rentals? • Should we measure total customer spending? • Should support tickets be included as part of customer activity?
Dimensions	• Should customer activity be analyzed by country and city? • Should activity be compared between subscription plans? • Should customers be analyzed by status or other customer groups?
Time Granularity	• Should customer activity be analyzed daily, weekly, monthly, or yearly? • What should one analytical record represent?
Reporting Frequency	• How frequently should customer activity reports be refreshed? • Is daily refresh sufficient or is near-real-time reporting required?
Historical Data	• How much historical customer activity should be retained? • Should previous customer attributes, such as location or subscription plan, be preserved?
Data Quality	• How should duplicate customer records across operational systems be handled? • How should missing customer information be handled? • Should inactive customers be included in the analysis?
Performance	• How quickly should customer activity reports return results? • What level of data volume is expected as the number of customers increases?

Security	• Which users should have access to customer-level data? • Should personally identifiable information such as email, phone number, and address be restricted?
Future Scalability	• Should the model support customers from multiple countries? • Should it support new subscription plans and customer channels in the future?

3.1.1 Assumptions and Justifications

Assumption	Justification
Customer activity includes both streaming and physical rental activities.	ABC Hub operates both digital streaming and physical rental services, and management wants to analyses customer behavior across both channels.
Customer activity will be analyzed mainly at customer-day level.	Daily data provides sufficient detail for trend analysis while allowing efficient aggregation into weekly and monthly reports.
Customer location will be analyzed by country and city.	ABC Hub plans to expand internationally, so geographical analysis will be important.
Customer activity data will be refreshed daily.	The main requirement focuses on BI and management reporting rather than real-time operational decisions.
Historical customer activity will be retained for long-term analysis.	Historical customer behavior is required for trend analysis and future machine-learning models for data scientists.

3.1.2 Refined Requirement

The analytical platform shall provide integrated customer activity analysis across both streaming and physical rental services.

Customer activity will be analyzed at customer-day level and will include measures such as:

- Number of streaming sessions
- Total streaming duration
- Number of physical rentals
- Number of returned items
- Total amount spent

- Number of support tickets raised

The data should support analysis by:

- Date
- Customer
- Country
- City
- Subscription plan

The dataset will be refreshed daily and historical customer activity will be retained for long-term trend analysis and future machine-learning use.

3.2 Streaming Performance

Initial Requirement: -Management wants to analyses streaming performance.

Requirement Area	Stakeholder Questions
Business Objective	• What decisions should streaming performance analysis support? • Is the objective to identify popular content, customer engagement, or streaming usage trends?
KPIs and Measures	• Should the number of streaming sessions be measured? • Should total watch duration be measured? • Should average completion percentage be calculated? • Should the number of unique viewers be measured?
Dimensions	• Should streaming performance be analyzed by content? • Should it be analyzed by content type or genre? • Should performance be compared by customer location? • Should device type be included?
Time Granularity	• Should streaming performance be analysed daily, weekly, or monthly? • Should detailed individual streaming sessions also be retained?
Reporting Frequency	• How frequently should streaming performance data be updated?
Historical Data	• How much historical streaming data should be retained? • Is historical streaming activity required for recommendation models?
Data Quality	• How should incomplete streaming sessions be handled? • What should happen if stream start or end timestamps are missing?

	• How should duplicate streaming sessions be identified?
Performance	• How quickly should reports for high-volume streaming data return results?
Security	• Should analysts have access to individual customer streaming behavior?
Future Scalability	• How much growth is expected in streaming sessions? • Should new device types and content types be supported?

3.2.1 Assumptions and Justifications

Assumption	Justification
Streaming performance will be analyzed daily and monthly.	Daily analysis supports detailed usage trends while monthly analysis supports management reporting.
Important measures include stream count, watch duration, completion percentage, and unique viewers.	These measures provide useful information about content usage and customer engagement.
Streaming data will be analyzed by content, content type, device, date, and location.	These dimensions allow management to understand how and where content is consumed.
Historical streaming data will be retained.	The Data Science team requires historical content usage information for future recommendation models.

3.2.2 Refined Analytical Requirement

The analytical platform shall support daily and monthly streaming performance analysis.

The main measures will include:

- Total streaming sessions
- Total watch duration
- Average completion percentage
- Unique viewers

Streaming performance should be analyzed by:

- Date
- Content
- Content type

- Genre
- Device
- Customer location

Historical streaming data will be retained to support trend analysis and future recommendation models.

3.3 Physical Rental Trends

Requirement Area	Stakeholder Questions
Business Objective	• What business decisions should rental trend analysis support? • Is management interested in rental demand, return behavior, or revenue?
KPIs and Measures	• Should the number of rentals be measured? • Should returned items be measured? • Should late returns be measured? • Should rental fees and late fees be included?
Dimensions	• Should rental trends be analyzed by content? • Should rentals be analyzed by warehouse or customer location? • Should content type and genre be included?
Time Granularity	• Should rental trends be analyzed daily, weekly, or monthly?
Reporting Frequency	• How frequently should rental reports be updated?
Historical Data	• How many years of rental history should be retained?
Data Quality	• How should missing return dates be handled? • How should cancelled rentals be treated? • How should invalid rental statuses be handled?
Performance	• How quickly should rental trend reports be generated?
Security	• Who should have access to customer-level rental information?
Future Scalability	• Should the model support additional warehouses and countries?

3.3.1 Assumptions and Justifications

Assumption	Justification
Rental trends will be analyzed daily and monthly.	This supports both detailed operational trends and management-level reporting.

Rental measures include rental count, return count, rental fees, and late fees.	These measures describe rental activity and financial performance.
Rental data will be analyzed by content, warehouse, location, and date.	These dimensions help identify demand and utilization patterns.
Historical rental records will be retained.	Historical information is required to identify long-term rental trends.

3.3.2 Refined Analytical Requirement

The analytical platform shall support physical rental trend analysis using historical rental information.

The main measures will include:

- Number of rentals
- Number of returns
- Rental revenue
- Late-fee revenue

Analysis will be available by:

- Date
- Content
- Content type
- Warehouse
- Country
- City

Historical rental information will be retained for long-term trend analysis.

3.4 Revenue

Requirement Area	Stakeholder Questions
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Business Objective	• What business decisions should revenue analysis support? • Should streaming subscriptions and physical rentals be compared?
KPIs and Measures	• What should be considered revenue? • Should subscription payments be included? • Should rental fees be included? • Should late fees be included? • Should failed payments be excluded? • How should refunded payments be treated?
Dimensions	• Should revenue be analyzed by country? • Should revenue be analyzed by subscription plan? • Should revenue be analyzed by payment method? • Should revenue be analyzed by content or rental type?
Time Granularity	• Is revenue analysis required daily, monthly, quarterly, or yearly?
Reporting Frequency	• How frequently should revenue information be refreshed?
Historical Data	• How many years of financial information should be retained?
Data Quality	• How should failed, cancelled, or duplicated payments be handled? • How should missing payment amounts be treated?
Performance	• What is the expected response time for financial reports?
Security	• Which roles should have access to detailed revenue and payment data?
Future Scalability	• Will revenue need to support multiple currencies? • Should financial reports support multiple countries and regions?

3.4.1 Assumptions and Justifications

Assumption	Justification
Revenue includes successful subscription and rental-related payments.	ABC Hub generates income mainly from subscriptions and physical rentals.

Failed and cancelled payments will not be counted as revenue.	These payments do not represent successfully received income.
Revenue will be analysed daily, monthly, and yearly.	These levels support operational, management, and long-term trend reporting.
Country and currency information will be supported.	International expansion may introduce multiple currencies and regional reporting requirements.

3.4.2 Refined Analytical Requirement

The analytical platform shall provide revenue analysis across subscription and physical rental services.

Revenue calculations will include valid successful payments and exclude failed or cancelled payments.

Revenue should be analysed by:

- Date
- Country
- Customer
- Subscription plan
- Payment type
- Payment method

The analytical platform should support future multi-country and multi-currency reporting.

3.5 Content popularity

Requirement Area	Stakeholder Questions
Business Objective	• How will content popularity information be used? • Should digital and physical content popularity be compared?
KPIs and Measures	• Should popularity be based on number of streams? • Should rental count be included? • Should customer ratings be included? • Should Wishlist additions be included? • Should unique viewers or customers be considered?
Dimensions	• Should popularity be analyzed by content type?

	• Should it be analyzed by genre? • Should it be analyzed by country or city? • Should release year or language be included?
Time Granularity	• Should content popularity be analyzed daily, weekly, or monthly?
Reporting Frequency	• How frequently should content popularity reports be updated?
Historical Data	• Should historical popularity rankings be retained?
Data Quality	• How should invalid ratings or duplicate Wishlist records be handled?
Performance	• How quickly should top-content reports be generated?
Security	• Who should have access to content-performance information?
Future Scalability	• Should new content types and genres be supported?

3.5.1 Assumptions and Justifications

Assumption	Justification
Content popularity will consider both digital streaming and physical rentals.	ABC Hub provides the same overall business through digital and physical channels.
Popularity will be measured using streams, rentals, ratings, and Wishlist additions.	A combination of these measures gives a more complete view of customer interest.
Content popularity will mainly be analyzed monthly.	Monthly aggregation is appropriate for identifying medium- and long-term popularity trends.

3.5.2 Refined Analytical Requirement

The analytical platform shall provide monthly content popularity analysis across both streaming and physical rental channels.

Measures will include:

- Total streams
- Total rental count
- Average customer rating
- Number of wishlist additions

Content popularity should be analysed by:

- Content

- Content type
- Genre
- Language
- Country
- Month

Historical content performance will be retained to support trend analysis and recommendation models.

3.6 Inventory Utilization

Requirement Area	Stakeholder Questions
Business Objective	• What decisions should inventory utilization analysis support? • Is the goal to identify underutilized or highly demanded inventory?
KPIs and Measures	• Should number of rentals be measured? • Should number of returns be measured? • Should available days be measured? • How should inventory utilization percentage be calculated?
Dimensions	• Should inventory utilization be analyzed by warehouse? • Should it be analyzed by content? • Should item condition be considered? • Should geographical location be included?
Time Granularity	• Should utilization be calculated daily, weekly, or monthly?
Reporting Frequency	• How frequently should inventory utilization information be refreshed?
Historical Data	• Should historically inventory status changes be retained?
Data Quality	• How should missing inventory status values be handled? • How should damage or unavailable inventory be treated?
Performance	• How quickly should inventory reports return results?
Security	• Which teams should have access to warehouse-level inventory information?
Future Scalability	• Should the solution support additional warehouses in new countries?

3.6.1 Assumptions and Justifications

Assumption	Justification
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Inventory utilization will be measured at inventory-item-per-day level.	This allows detailed utilization analysis and supports aggregation to warehouse and monthly levels.
Important measures include rental count, return count, available days, and utilization percentage.	These measures describe how effectively physical inventory is being used.
Inventory utilization will be analyzed by warehouse, content, and date.	These dimensions help management identify demand and inventory distribution problems.

3.6.2 Refined Analytical Requirement

The analytical platform shall provide daily inventory utilization analysis for each physical inventory item.

Measures will include:

- Number of rentals
- Number of returns
- Number of days available
- Inventory utilization percentage

Analysis should be available by:

- Inventory item
- Content
- Warehouse
- City
- Country
- Date

Historical inventory information will be retained to support capacity planning and future warehouse expansion.

4 Historical Data and Traceability with Data Vault

The company needs to retain complete historical information and provide full data traceability for auditing and regulatory purposes.

To achieve that I suggest **Data vault data modeling** approach in our **silver layer** in medallion architecture.

Data Vault separates data into **Hubs, Links, and Satellites**.

- Hubs store business keys
- Links store relationships between business entities.
- Satellites store descriptive attributes and their historical changes.

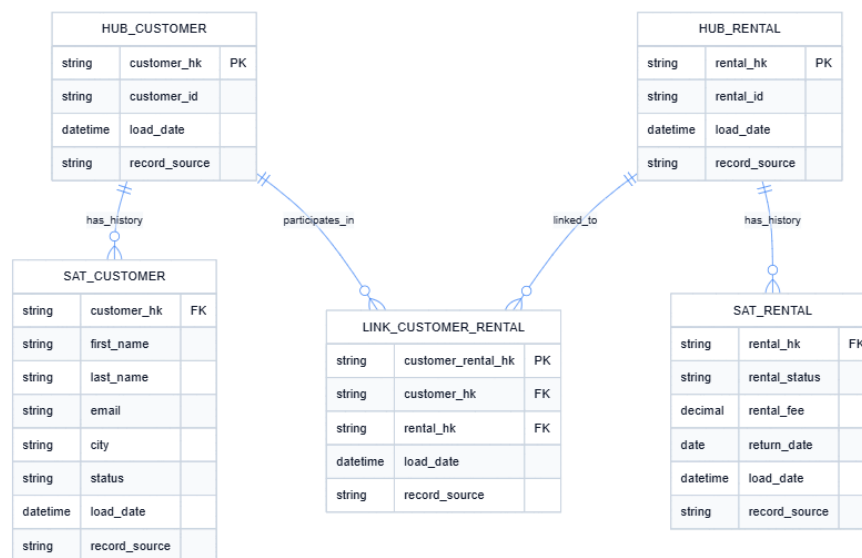


Figure 3 Data Vault Model for Historical Data Retention and Traceability

Hubs such as HUB_CUSTOMER and HUB_RENTAL store stable business keys. This allows business entities to be uniquely identified even when their descriptive information changes.

Links such as LINK_CUSTOMER_RENTAL store relationships between business entities. For example, the relationship between a customer and a rental can be preserved independently from the descriptive information.

Satellites such as SAT_CUSTOMER and SAT_RENTAL store descriptive attributes and their historical changes. When a customer's city, status, or other attribute changes, a new Satellite record can be added instead of overwriting the previous value. Therefore, complete historical information can be retained.

The diagram also shows metadata such as **load date** and **record source**. These fields identify **when the data was loaded and where it came from**, which provides traceability for auditing purposes.

When data changes, the old value is not overwritten. A new Satellite record is added, which helps preserve historical data.

Data Vault also stores metadata such as **load date** and **record source**, which helps identify when the data was loaded and where it came from. This will support **auditing and regulatory requirements**.

5 Analytical Modeling with Kimball

The company needs to prepare reports and perform analysis efficiently. This can be achieved by applying **Kimball dimensional modelling in the Gold layer**.

Instead of asking analysts to join many operational tables, the Gold layer will provide simple **Star Schemas** made of Fact tables and Dimension tables.

The dimensional model improves performance because:

- Analysts need fewer joins.
- Fact tables store ready-to-use business measures.
- Dimension tables make filtering and grouping simple.
- Aggregated fact tables can reduce the amount of data scanned.
- Indexes and partitioning can be applied to analytical tables.
- BI tools can query the Gold layer directly instead of complex source structures.

Advantages

- Easy for business analysts to understand.
- Faster report generation.
- Consistent KPI definitions.
- Good support for BI tools.
- Supports self-service analytics.
- Reduces dependency on IT teams.

Limitations / Trade-offs

- Some data is duplicated between dimension and fact structures.
- ETL logic becomes more complex because data must be transformed from the Silver layer into the Gold layer.
- Changes in business requirements may require updates to Fact or Dimension tables.
- Dimensional models are not ideal for full historical traceability, so the Data Vault in the Silver layer should still be maintained.

6 End-to-End Modern Data Platform Architecture

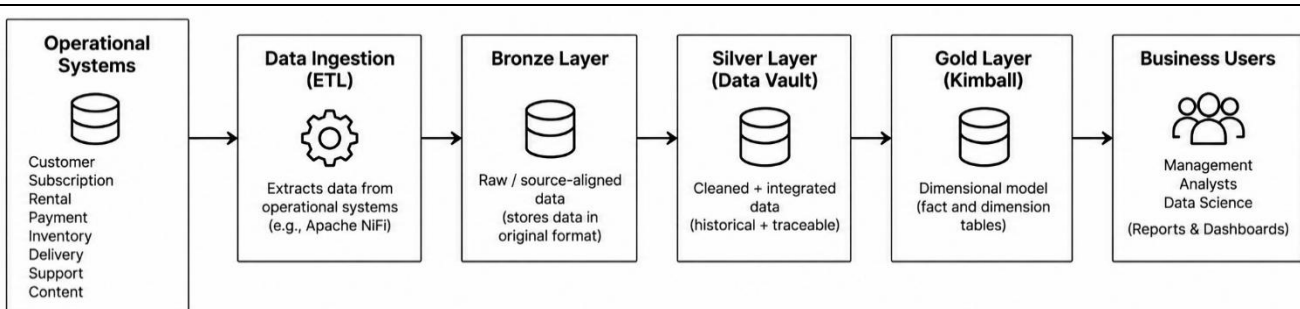


Figure 4 End-to-End Modern Data Platform

7 Dimensional Models and Star Schemas



Figure 5 Customer Daily Activity Star Schema

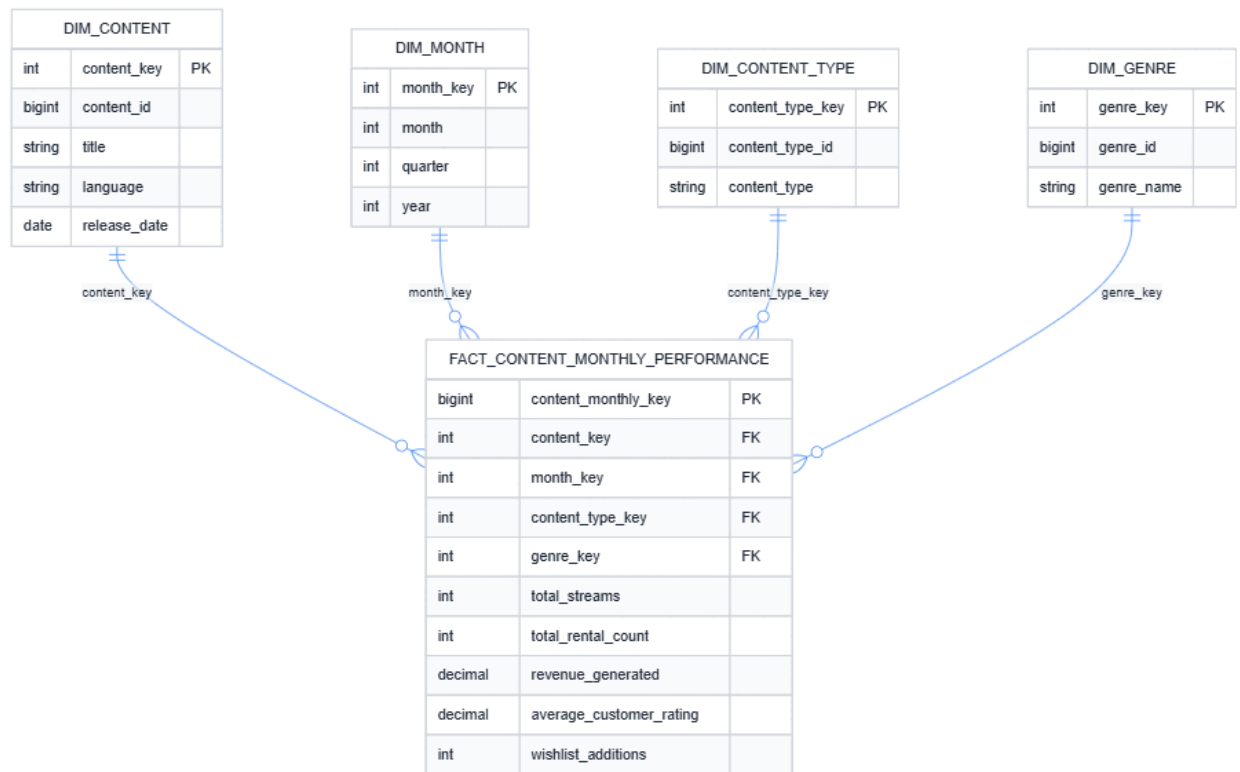


Figure 6 Content Monthly Performance Star Schema



Figure 7 Inventory Utilization Star Schema

8 ETL Implementation with Apache NiFi

8.1 Overall ETL Pipeline

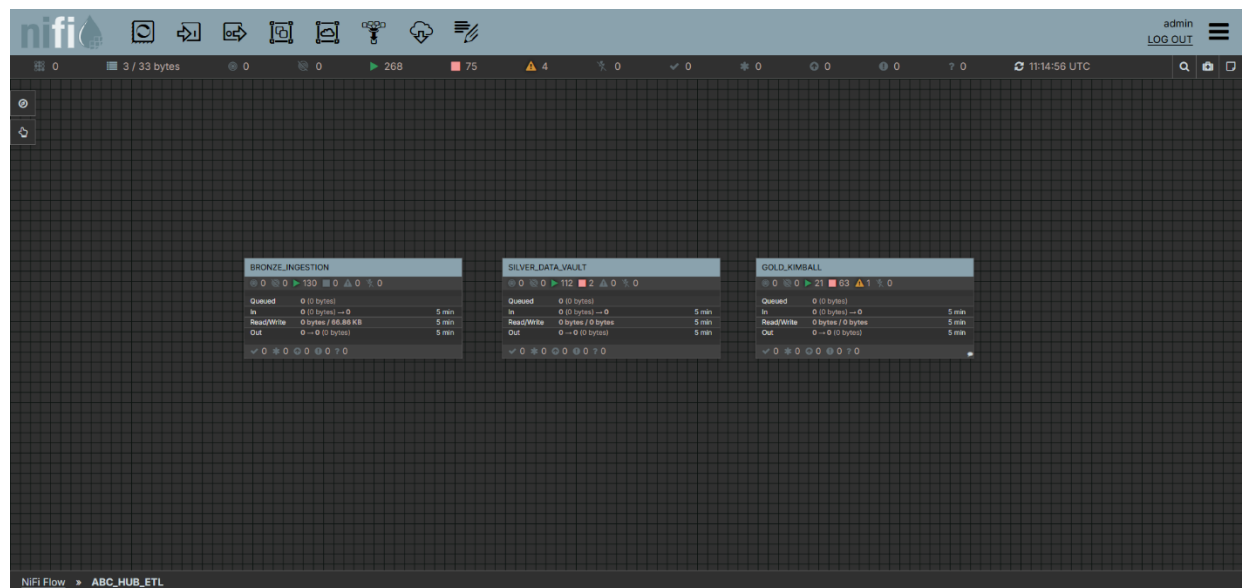


Figure 8 Overall Apache NiFi ETL Pipeline

8.2 Bronze Layer Ingestion

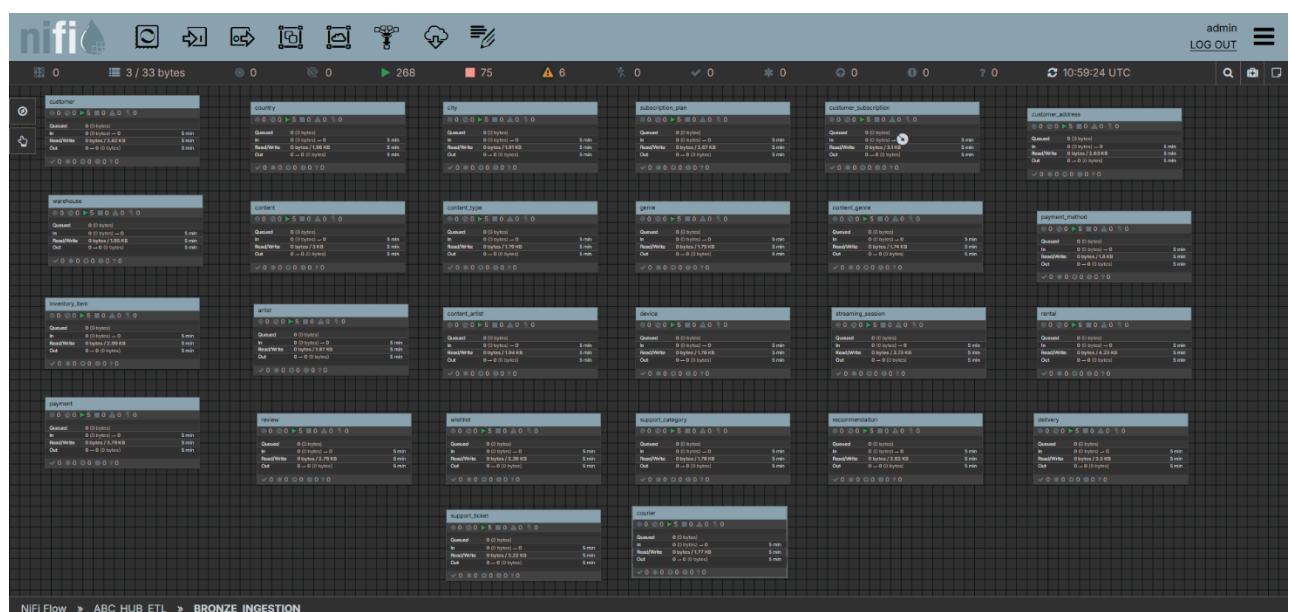


Figure 9 Bronze Layer Ingestion Flow

8.3 Silver Data Vault Processing

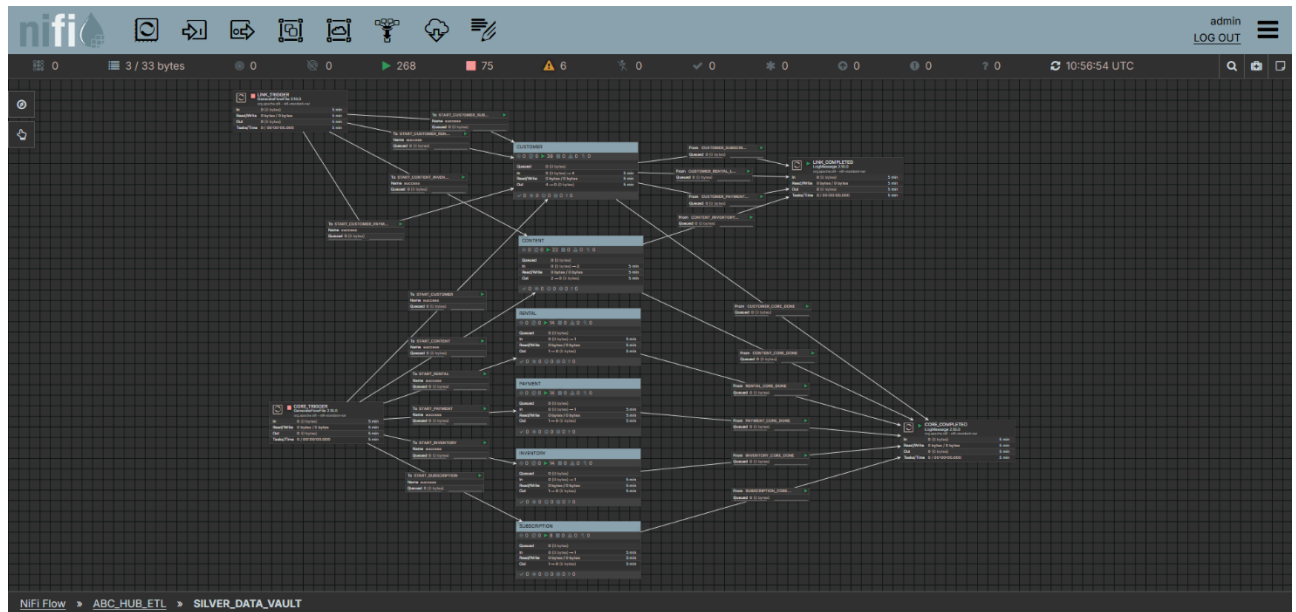


Figure 10 Silver Data Vault Processing Flow

8.4 Gold Kimball Processing

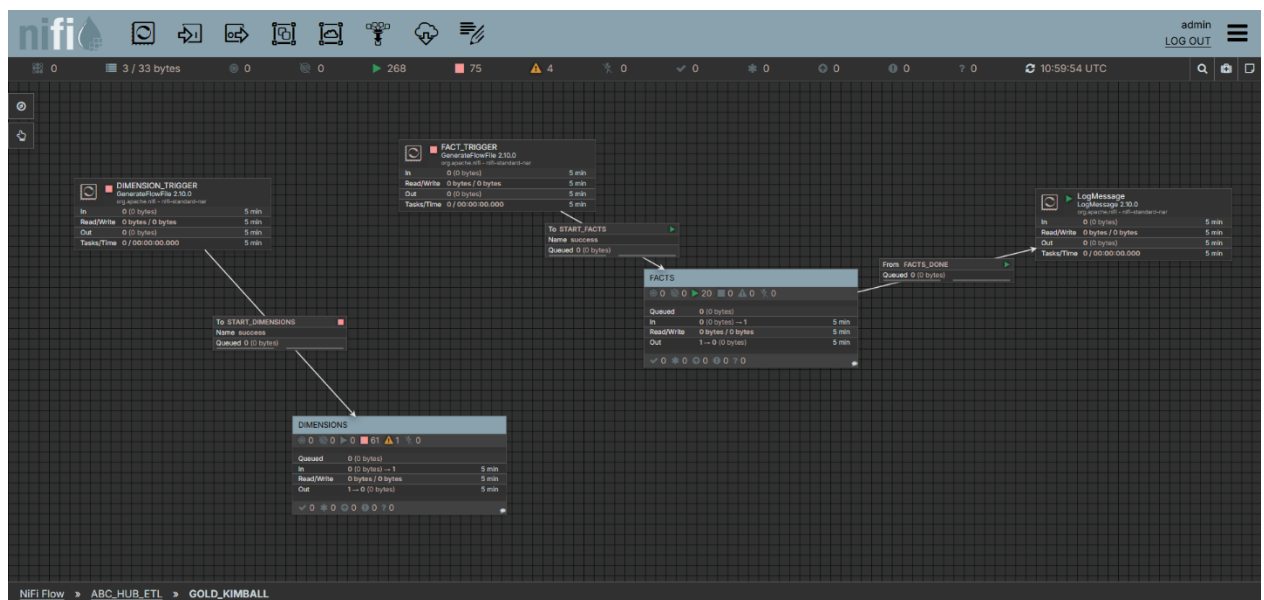


Figure 11 Gold Kimball Dimension and Fact Loading Flow